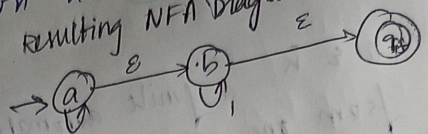
What strings does it accept?

the automation accepts strings of the form.

0^*1^*

Resulting NFA Diagram:



this means:

→ Zero or more 0s, followed by zero or more 1s

→ All the 0s must come before the 1s.

→ The empty string ε is also accepted.

Examples:

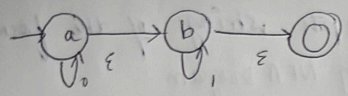
I/P string	Result	Reason
0	Accept	Read 0, then move to the final state using ε-moves
ε	Accept	can reach the final state using ε-moves
(1)	Accept	move from a to b using ε, read 1, then reach the final state

Language accepted:

$L = \{0^m 1^n \mid m, n \geq 0\}$

Type: NFA with ε-transitions (NFA-ε)

FINITE AUTOMATA WITH NULL MOVES (NFA-ε)



An NFA-ε is a finite automation that allows transitions from one state to another without reading any input symbol. These transitions are called epsilon(ε) transition.

Eg: In the given diagram, the automation can move from state a to state b without reading any input using an ε - transition.

Given,

State a → Initial State.
State b → Intermediate state.
Double circle → Final / accepting state.
ε(epsilon) → Moves to another state
Loop on a labelled 0 → Reads 0 & stays in state a
Loop on b labelled 1 → Reads 1 & stays in state b.

Current State	Input	Next state
a	0	a
a	ε	b
b	1	b
b	ε	Final state (qf)